

Parametric Adaptation Methods for Document-Grounded Legal QA

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Abstract

Parameter-efficient adaptation may improve document-grounded legal QA, but its value is unclear when Retrieval-Augmented Generation (RAG) already supplies the source documents. We compare RAFT-style supervised fine-tuning with Causal Language Modeling (CLM) continued pretraining on the same frozen language model and DIFC benchmark, holding retrieval fixed.

Retrieval-aware systems outperform closed-book controls, indicating that adaptation changes evidence-conditioned answer behavior rather than replacing retrieval. RAFT improves deterministic extraction, whereas CLM improves free-text synthesis and explanation quality. A post-hoc adapter merge recovers RAFT’s deterministic advantage and yields the highest multi-document score, but has higher seed variance and does not retain CLM’s free-text edge. Under hardware constraints, the training signal should match the required answer profile. Adapter fusion remains exploratory.

Keywords: Retrieval-augmented generation, Parameter-efficient fine-tuning, QLoRA, Continued pretraining, Domain adaptation, Legal question answering

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1 Introduction

1.1 Problem and Motivation

Document-grounded legal question answering requires both factual precision and answer discipline. The target answer is frequently tied to a specific provision, date, list element, or procedural distinction, so unsupported generation is costly. Improvements should be credited when they strengthen document-bound answering under fixed evaluation conditions.

Resource limits make scaling to larger language models impractical in this setup and shift the design space toward retrieval engineering, smaller generators, and parameter-efficient adaptation. Practitioners then face a concrete engineering choice: invest in retrieval, adapt the generator, or combine both.

The central question is whether adapting the generator adds value once retrieval is already in place, and whether different adaptation signals produce different quality profiles under identical infrastructure constraints.

1.2 Research Questions and Scope

The scope is deliberately narrow: one legal benchmark, one frozen language model, one hardware configuration, and one fixed retrieval design. Two research questions guide the investigation:

RQ1. Does parametric adaptation yield gains over the RAG baseline on a compact legal benchmark under tight resource constraints, and how do RAFT-style supervised adaptation and supervision-free CLM continued pretraining differ as retrieval-conditioned generators?

RQ2. How far can pure parametric systems reach without retrieval on this benchmark, and does retrieval remain necessary?

The small corpus and evaluation split limit external validity; the findings apply only to the evaluated benchmark and system configuration.

1.3 Contributions

The paper makes three empirical contributions:

1. A comparison between RAFT-style supervised adaptation and CLM continued pretraining using the same language model, retrieval settings, and PEFT architecture.
2. A quantification of the limits of pure parametric memory by contrasting retrieval-aware systems against no-retrieval controls, thereby clarifying whether retrieval remains necessary in this setting.
3. A post-hoc adapter-merge result suggesting partial complementarity between the two training signals, reported as an exploratory finding that remains secondary to the headline comparison.

1.4 Structure of the Paper

Section 2 provides background on RAG, parameter-efficient adaptation, and the two training paradigms under comparison. Section 3 describes the benchmark corpus, hardware constraints, and the fixed retrieval configuration. Section 4 defines the compared systems and the evaluation protocol. Section 5 presents the experimental results, including aggregate comparisons, per-type analyses, and a single-document versus multi-document breakdown. Section 6 discusses the findings in light of the research questions, analyzes common error patterns, and acknowledges the limitations. Section 7 concludes with practical implications and directions for future work.

2 Background and Related Work

2.1 RAG as Nonparametric Memory

Retrieval-augmented generation (RAG) supplies a generator with passages retrieved from an external corpus (Lewis et al., 2020). In legal QA, this supports page-level evidence tracing and gives a small model access to document content without requiring it to memorize the corpus. Parametric adaptation is evaluated here on top of the same hybrid retrieval, reranking, and evidence-compression setup, so gains are measured relative to an existing external-memory mechanism.

2.2 Parameter-Efficient Adaptation under Resource Constraints

Full fine-tuning stores optimizer states and gradients for every parameter, exceeding the available memory in this setting. Low-Rank Adaptation (LoRA) addresses this by freezing the pretrained weights and injecting small trainable rank-decomposition matrices into selected attention layers (Hu et al., 2022). QLoRA extends this approach by quantizing the frozen model to 4-bit NormalFloat (NF4) precision, reducing memory consumption further while preserving adaptation quality (Dettmers et al., 2023). Survey and review

work consistently treats this family of methods as a quality-versus-resource trade-off (Han et al., 2024; Xu et al., 2023).

QLoRA makes adaptation of the shared language model feasible within the available memory. Both adapted systems use the same QLoRA configuration, so they differ in training signal rather than adaptation mechanism.

2.3 RAFT-style Adaptation vs. CLM Continued Pretraining

The central experimental axis contrasts two training signals applied to the same PEFT architecture.

RAFT-style supervised adaptation. Inspired by Retrieval-Augmented Fine-Tuning (Zhang et al., 2024), the adapter is trained on question-answer pairs where the input includes retrieved evidence chunks (both gold and distractor passages). This directly optimizes answer generation from evidence-rich contexts, exposing the adapter to the QA task distribution. The training signal is supervised: labels are the reference answers.

CLM continued pretraining. The adapter is trained on the raw corpus text using a standard causal language modeling (CLM) objective - next-token prediction on all tokens. No QA labels or task-specific formatting are used. The adapter is exposed to the corpus distribution without any task-specific supervision, relying solely on the language modeling objective to absorb domain patterns.

These two paradigms represent different assumptions about how parametric adaptation should interact with retrieval. RAFT-style training teaches the generator *how to use* retrieved evidence, whereas CLM pretraining teaches it *what the corpus contains*. RAFT-style adaptation may favor deterministic extraction because it is trained directly on answer production under evidence conditioning. CLM adaptation may favor assistant-style answer quality because it changes the model’s local contextualization behavior without task-specific labels. The empirical question is which of these tendencies becomes visible once both are tested against the same RAG baseline.

2.4 Research Gap and Positioning

High-level discussions of parametric versus nonparametric knowledge injection are abundant in the literature. In the legal domain specifically, benchmarks such as LegalBench (Guha et al., 2023), LegalBench-RAG (Pipitone & Alami, 2024), and evaluation frameworks like LRAGE (Park et al., 2025) have evaluated LLM capabilities for legal reasoning and retrieval-augmented legal QA. However, few studies compare training signals while holding the retrieval infrastructure, PEFT architecture, language model, and evaluation protocol fixed. Most comparisons involve different model families, retrieval setups, or evaluation protocols, making it difficult to attribute performance differences to the training signal alone.

The comparison addresses legal QA on resource-constrained hardware. It does not claim that any method is universally best; on this benchmark, adaptation yields moderate gains over the RAG baseline, and those gains depend on the training signal rather than on the mere presence of an adapter.

3 Benchmark and Experimental Setup

3.1 Corpus and Benchmark

The benchmark is built on 8 PDF documents from the DIFC legal corpus, comprising statutes, regulations, and court judgments. Together, the documents span approximately 176 pages and 115,000 tokens. A pool of 200 question–answer pairs was authored by domain experts, covering six answer types: free-text explanations (53 questions), boolean lookups (48), numeric extractions (36), named entity lookups (30), multi-name lists (17), and date extractions (16). The distribution includes 26 multi-document comparative questions (13%) and 17 unanswerable questions (8.5%), ensuring that evaluation is not limited to simple single-document lookups. Difficulty labels span easy, medium, and hard cases.

The benchmark combines heterogeneous answer types, from boolean lookups to free-text legal explanations. This heterogeneity exposes distinct failure modes and prevents aggregate scores from masking type-specific weaknesses. The multi-document subset additionally provides a natural stress test for systems that may differ in local contextualization versus cross-document aggregation.

The 200 questions are split into 150 training questions and 50 evaluation questions, stratified by answer type, difficulty, and single-/multi-document status. This frozen split is used throughout the paper: all systems are evaluated on the identical 50-question evaluation set. Supervised training (RAFT-style adaptation) uses only the 150 training questions; CLM continued pretraining uses the raw document text and is therefore independent of the QA split. The small benchmark limits generalizability and statistical power, particularly for answer types with few evaluation items.

3.2 Hardware, Shared Model, and Variance Policy

All experiments run on a single NVIDIA RTX 4060 with 8 GB VRAM and 32 GB system RAM. The shared language model is Gemma-2-2b-it, an instruction-tuned model with approximately 2 billion parameters. It is held constant across all systems to prevent architectural variance from confounding the training-signal comparison: the headline systems differ in adaptation signal, not in model family or deployment environment.

For systems that involve training (RAFT-RAG, CLM-RAG, and their no-retrieval controls), three random seeds (42, 123, 777) are used, and results are reported as mean $\pm$ standard deviation. No cross-validation is performed: the single frozen split is shared across all evaluations, and seed-level variance captures only the stochasticity introduced by the training process. This variance policy is modest, but it provides a clearer view of stability than a single run would, while keeping all compared systems anchored to the same test set.

3.3 Fixed Retrieval Configuration

The retrieval configuration is held constant across all retrieval-aware systems (Base-RAG, RAFT-RAG, CLM-RAG, Merge-RAG). It comprises five stages:

1. **Ingestion and hierarchical chunking.** Documents are parsed and split into five chunk families: page-level, section-level, clause-level, microchunks (300 tokens, 50-token overlap), and table blocks. Metadata - including entities, dates, heading paths, and BM25 terms - is extracted for each chunk.
2. **Hybrid retrieval.** Each query is embedded using Qwen3-Embedding-0.6B (384 dimensions) for dense retrieval and tokenized for BM25 sparse retrieval ($k_1 = 1.5$, $b = 0.75$). Both channels prefetch 30 candidates.
3. **Reciprocal Rank Fusion (RRF).** Dense and sparse candidate lists are fused with equal weights and $k = 60$, producing a ranked list of 10 candidates.
4. **Cross-encoder reranking.** The top 10 candidates are reranked using Qwen3-Reranker-0.6B, and the top 5 are retained.

5. **Evidence compression.** A page-diverse compressor selects up to 3 chunks (at most one per physical page), and the corresponding (doc_id, page_number) pairs are lifted for grounding evaluation.

System differences are therefore interpreted at the generator stage. Exact retrieval parameters are listed in Appendix A.

4 Compared Systems and Evaluation Protocol

4.1 System Inventory

Seven systems occupy distinct methodological roles. Three form the headline comparison, one provides an exploratory post-hoc result, and three serve as negative controls. Table 1 summarizes their key characteristics.

Table 1: Compared systems and their roles.

System	Retrieval	Training signal	Supervision	Role
Base-RAG	Yes	None	—	Headline baseline
RAFT-RAG	Yes	RAFT-style QA	Supervised	Headline
CLM-RAG	Yes	CLM on corpus	Unsupervised	Headline
Merge-RAG	Yes	Merged RAFT + CLM	Post-hoc	Exploratory
RAFT-Closed	No	RAFT-style QA	Supervised	Control
CLM-Closed	No	CLM on corpus	Unsupervised	Control
D2L-Closed	No	D2L hypernetwork	Supervised	Control

Base-RAG serves as the nonparametric baseline: the frozen Gemma-2-2b-it generator receives retrieved evidence and produces answers without any adapter.

RAFT-RAG and **CLM-RAG** are the two headline adapted systems. Both receive the same retrieved evidence as the baseline and use the same QLoRA architecture (rank 32, alpha 32, dropout 0.05, targeting q_proj and v_proj). Their training recipes differ: RAFT-RAG uses

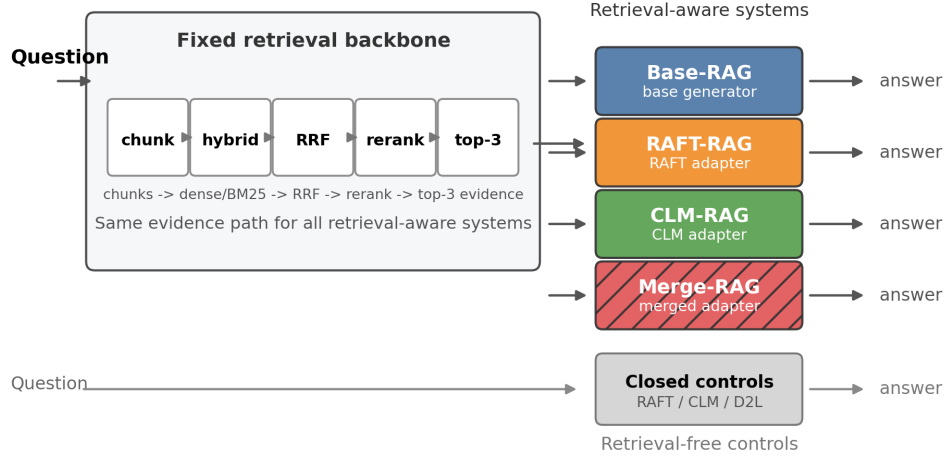


Figure 1: System overview schematic. Base-RAG routes queries through the shared retrieval stack to the base generator; RAFT-RAG and CLM-RAG route through retrieval to an adapted generator; Merge-RAG uses a merged adapter; controls bypass retrieval entirely.

question-answer pairs with retrieved context, whereas CLM-RAG uses raw corpus text and causal language modeling. These three systems define the main thesis comparison.

Merge-RAG (Post-hoc adapter fusion) linearly interpolates the RAFT-RAG and CLM-RAG adapters with equal weights ($\alpha = 0.5$), pairing source adapters by matching training seed, without any additional training, and receives the same retrieved evidence. Because Merge-RAG inherits prior training effort and is not a separately trained system, it is reported outside the headline branch as an exploratory result.

RAFT-Closed is trained separately on question-answer pairs without retrieved context. **CLM-Closed** reuses the CLM-RAG adapter but bypasses retrieval at inference time. These controls clarify the limits of parametric memory without retrieval and are not part of the main claim. **D2L-Closed** is a secondary control using a Doc-to-LoRA hypernetwork approach (Charakorn et al., 2026). The released implementation imposes a per-adapter token limit, while adaptation to a modern target model would require a bespoke hypernetwork that was not trained here. D2L-Closed is therefore reported as an engineering control, with diagnostics in Appendix C.

4.2 Training Setups

Both RAFT-RAG and CLM-RAG employ identical QLoRA architectures applied to the same frozen model. Their complete training recipes differ in objective, data, learning rate, epoch count, and maximum sequence length. The comparison therefore evaluates the two recipes rather than isolating the training objective alone.

RAFT-RAG training. The adapter is fine-tuned for 3 epochs on the 150 training questions in RAFT format. Each training example consists of the question, gold evidence chunks (matched to gold retrieval pages), and 2 distractor chunks from unrelated documents. The target is the reference answer. Learning rate is 2×10^{-4} with cosine decay and 3% warmup. Maximum sequence length is 4096 tokens.

CLM-RAG training. The adapter is pretrained for 5 epochs on the concatenated corpus text (~115K tokens) using a CLM objective. Learning rate is 5×10^{-5} with cosine decay and 10% warmup. Maximum sequence length is limited to 512 tokens because longer sequences exceed the available memory at the logits stage. The same adapter is reused without retrieval as the CLM-Closed control.

RAFT-Closed control training. The closed-book supervised control is intentionally matched to RAFT-RAG in optimizer and PEFT settings (learning rate 2×10^{-4} , cosine schedule, 3% warmup, 3 epochs, maximum sequence length 4096). The training data format is the only difference: it omits retrieved context and uses question-to-answer pairs alone.

Merge-RAG. No training is performed. The RAFT-RAG and CLM-RAG adapter weight matrices are linearly interpolated per matching seed pair (42, 123, 777): $W_{\text{merged}} = 0.5 * W_{\text{RAFT-RAG}} + 0.5 * W_{\text{CLM-RAG}}$.

4.3 Evaluation Protocol

The evaluation protocol combines deterministic scoring for structured answer types with judge-based assessment for free-text responses, alongside grounding and operational metrics.

Composite metric. The primary metric is $Q_{\text{main}} = 0.7 * S_{\text{det}} + 0.3 * S_{\text{asst}}$, weighting deterministic extraction at 0.7 and judged free-text quality at 0.3. This weighting prioritizes factual precision while still crediting assistant-style quality on free-text answers.

Deterministic score (S_{det}). For boolean and date questions, scoring is binary exact match after normalization. Numeric answers are scored with exact match under a 1% tolerance. Single-name answers use normalized exact string match; multi-name lists are scored as the Jaccard similarity between predicted and gold name sets.

Unanswerable items. Unanswerable questions are handled differently depending on answer type. For deterministic unanswerable questions, the gold answer is null and the expected system output is the empty list `[]`; a system receives 1.0 only when it returns `[]`, and 0.0 otherwise. Free-text unanswerable questions remain part of the judged free-text subset, not of S_{det} : they are scored through the same judge procedure as other free-text answers, with the calibration criterion rewarding an explicit statement that the requested information is absent or unsupported.

Free-text score (S_{asst}). Free-text responses are evaluated by GPT-5.4-mini (OpenAI; model id `gpt-5.4-mini`, reasoning effort = medium), held fixed across all systems and experiments, against 5 binary criteria: correctness, completeness, grounding, calibration, and clarity (following the LLM-as-judge paradigm; see Pradhan et al., 2025 for a discussion of this approach in legal RAG evaluation). The per-question score is the mean of the 5 criteria; S_{asst} is the mean across all free-text questions. The judge prompt is frozen and identical for all systems (Appendix A.5). Malformed judge output is retried once; if the retry also fails, all five criteria are scored as zero for that answer. Before final interpretation, a manual audit of approximately 10% of judged free-text responses was performed, spot-checking judge scores against the rubric for systematic errors. Judge-based scoring is never used for deterministic answer types.

Grounding (G). For retrieval-aware systems, grounding is computed as F_{beta} (beta = 2.5) on page-level (`doc_id`, `page_number`) pairs, comparing the final evidence set against gold retrieval references. The elevated beta emphasizes recall, penalizing missing gold pages more than including extra pages. Because retrieval is fixed, grounding serves as a control on evidence access: the constant $G = 0.567$ across all retrieval-aware systems

indicates that the adapters change how the generator uses evidence, not which pages it receives.

Operational metrics. Latency (time-to-first-token and end-to-end), peak inference VRAM, offline training cost, and malformed output rate are reported for all systems; the full breakdown is given in Appendix B.3, and Table 2 summarizes the headline figures. Quality and resource expenditure are interpreted together, with direct offline-cost comparison restricted to systems that are genuinely comparable in training or packaging effort.

5 Results

5.1 Main Comparison

Table 2 presents the aggregate results across all systems. The headline systems are grouped at the top, followed by the exploratory post-hoc merge, and then the negative controls.

Table 2: Main results on the 50-question evaluation set. Trained systems report mean $\pm$ std across 3 seeds. Offline cost is per-seed wall-clock training time. Merge-RAG is excluded from direct offline-cost comparison because it inherits prior adaptation cost from both source adapters.

					Latency	VRAM	
System	Q_main	S_det	S_asst	G	(ms)	(MB)	Offline (s)
Headline							
Base-RAG	0.643	0.601	0.739	0.567	479	5201	—
RAFT-RAG	0.669 ± 0.014	0.648 ± 0.015	0.718 ± 0.018	0.567	492	3069	1206
CLM-RAG	0.667 ± 0.023	0.599 ± 0.016	0.826 ± 0.062	0.567	525	3069	581
Post-hoc							
Merge-RAG	0.705 ± 0.035	0.679 ± 0.048	0.764 ± 0.018	0.567	527	3069	n.c.*
Controls							
RAFT-Closed	0.263 ± 0.005	0.270	0.246	—	257	3067	88
CLM-Closed	0.185 ± 0.003	0.135	0.303	—	195	3077	581
D2L-Closed	0.210	0.135	0.385	—	179	3072	3932

* Not directly comparable: Merge-RAG inherits the combined offline cost of RAFT-RAG (1206 s) and CLM-RAG (581 s); the merge itself is instantaneous.

The RAG baseline (Base-RAG) reaches $Q_{\text{main}} = 0.643$, establishing a difficult starting point for any adapted retrieval-aware system. Both retrieval-aware adapters improve on it: RAFT-RAG attains 0.669 ± 0.014 and CLM-RAG attains 0.667 ± 0.023 . The observed improvements are moderate (+0.026 for RAFT-RAG and +0.025 for CLM-RAG) and consistent across seeds. No paired significance test or confidence interval over the 50 evaluation questions is reported, so these values remain observed differences rather than statistically established gains. The gains are small and measured with fixed evidence selection.

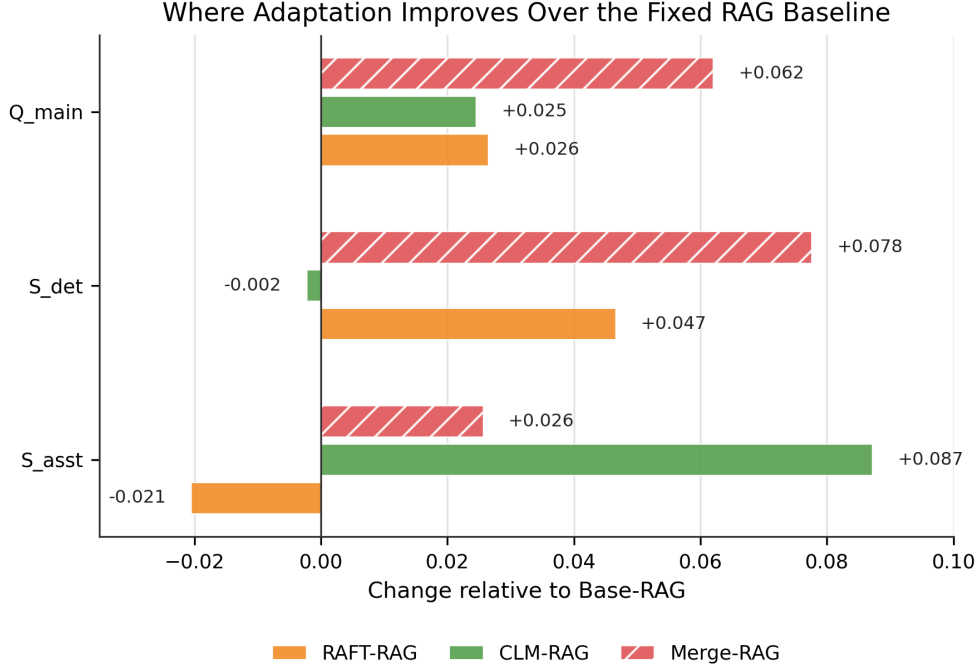


Figure 2: Improvement over Base-RAG. Grouped bars show Delta-Q_main, Delta-S_det, and Delta-S_asst for RAFT-RAG, CLM-RAG, and Merge-RAG: RAFT raises S_det, CLM raises S_asst, and the merge raises both.

Merge-RAG reaches the highest observed score (0.705 ± 0.035) through post-hoc adapter interpolation. However, its higher seed variance ($\text{std} = 0.035$ vs. 0.014 and 0.023 for the headline adapters) and post-hoc nature warrant cautious interpretation. It supports partial complementarity between the two adaptation signals, but it does not supersede the predefined headline comparison.

5.2 Trade-off Between RAFT-RAG and CLM-RAG

The headline comparison does not yield a single dominant system. Instead, the two adapters improve different quality dimensions, which is central to the analysis.

RAFT-RAG achieves higher deterministic extraction scores ($S_{\text{det}} = 0.648$ vs. 0.599), reflecting its supervised exposure to question-answer pairs with evidence context. CLM-RAG achieves substantially higher free-text answer quality ($S_{\text{asst}} = 0.826$ vs. 0.718), suggesting that CLM pretraining improves the generator’s ability to produce well-structured legal explanations. On the aggregate Q_main, the two systems are near-tied ($\text{delta} = 0.002$),

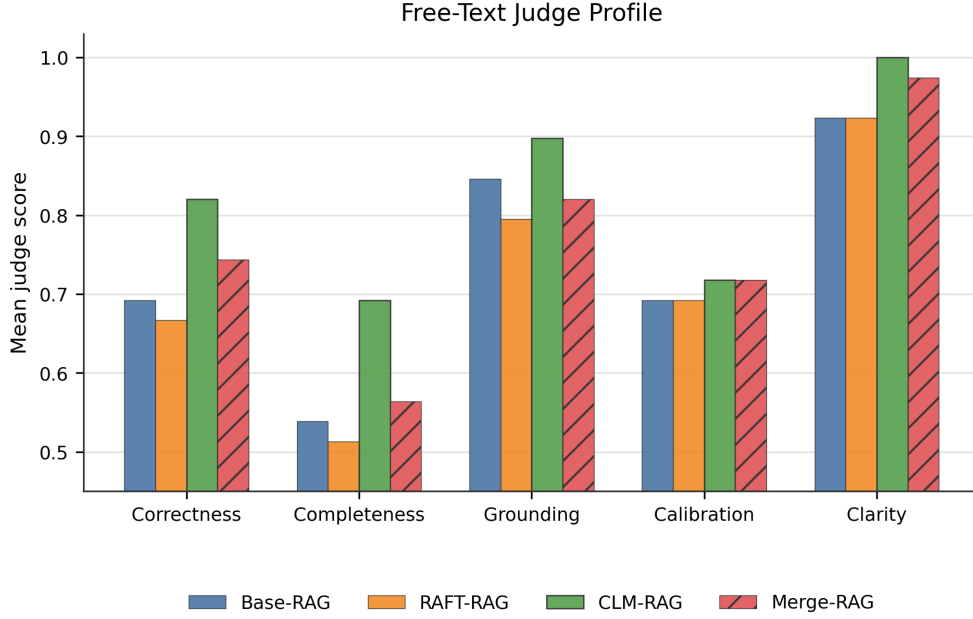


Figure 3: Judge criteria profile comparing Base-RAG, RAFT-RAG, CLM-RAG, and Merge-RAG on the 5 judge criteria (correctness, completeness, grounding, calibration, clarity). CLM-RAG’s advantage is concentrated in free-text quality dimensions.

with RAFT-RAG marginally ahead; the difference is too small to support a claim of practical dominance.

Relative to the RAG baseline, RAFT-RAG improves Q_{main} by +0.026 and S_{det} by +0.047 while slightly reducing S_{asst} by -0.021. CLM-RAG improves Q_{main} by +0.025 and S_{asst} by +0.087 while leaving S_{det} essentially unchanged (-0.002). This pattern supports the interpretation that training signal matters more than the mere presence of an adapter.

CLM-RAG also incurs lower offline cost (581 s vs. 1206 s per seed), as it requires no task-specific label generation. This matters under tight resource constraints, where training time competes with other workloads. The comparison thus records a tie on Q_{main} and grounding, with RAFT-RAG favored on deterministic extraction and CLM-RAG favored on assistant-style quality and offline cost.

5.3 By Answer Type

Type-level analysis reveals that performance differences between systems are concentrated in specific answer categories. Table 3 presents per-type scores broken down by the six answer types; each cell reports the metric appropriate to that type rather than the composite Q_main.

Table 3: Per-type scores on the 50-question evaluation set. Each cell is the type-appropriate score: S_det for the deterministic types (boolean, number, name, names, date) and S_asst for free-text. Headline and exploratory systems only; control systems are in Appendix B.

System	Number		Name	Names	Date (n=5)	Free-text (n=13)
	Boolean (n=12)	(n=7)	(n=8)	(n=5)		
Base-RAG	0.833	0.714	0.500	0.450	0.200	0.739
RAFT-RAG	0.889	0.714	0.625	0.261	0.400	0.718
CLM-RAG	0.833	0.714	0.583	0.300	0.200	0.826
Merge-RAG	0.889	0.810	0.708	0.224	0.400	0.764

The largest divergences appear between deterministic extraction and free-text explanation. RAFT-RAG outperforms the RAG baseline on boolean (+0.056), name (+0.125), and date (+0.200) types, consistent with its supervised training on structured answer extraction. CLM-RAG shows its advantage primarily on free-text (+0.087 relative to the baseline), where judged quality benefits from the CLM adapter’s exposure to corpus-level language patterns.

The breakdown also shows that no system performs uniformly well. The multi-name category remains difficult for the adapted systems, with both RAFT-RAG (0.261) and CLM-RAG (0.300) underperforming the RAG baseline (0.450). Date extraction remains weak across all systems: even the best system achieves only 0.400 on dates (n=5). These results point to persistent formatting and evidence-utilization limitations that neither training signal fully addresses. The main interpretive point is that near-equal aggregate scores conceal distinct answer behaviors that align with the different adaptation signals.

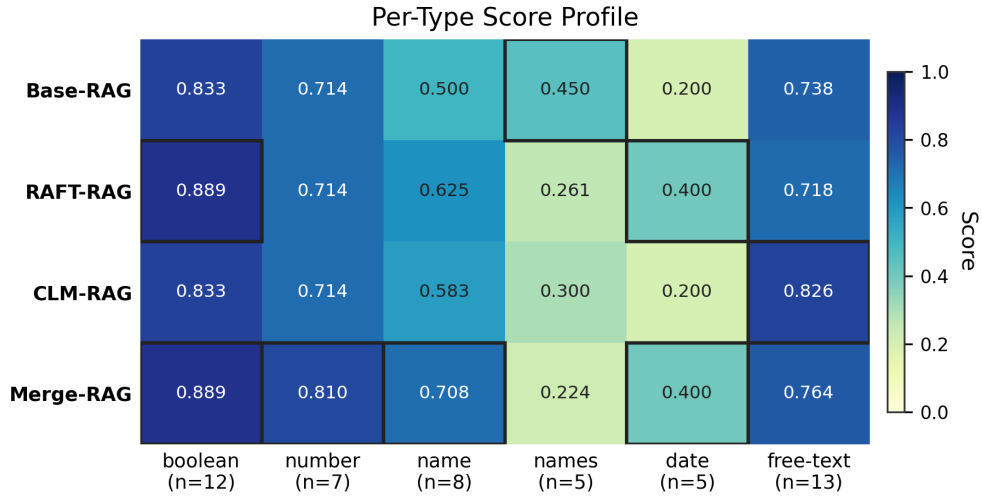


Figure 4: Per-type score heatmap for Base-RAG, RAFT-RAG, CLM-RAG, and Merge-RAG across the 6 answer types, with sample sizes in labels.

Merge-RAG achieves the highest score in 4 of 6 types, including number (0.810) and name (0.708), providing further evidence that the two training signals are partially complementary, though this observation remains secondary given the system’s post-hoc nature.

5.4 Retrieval Contribution and the Limits of Pure Parametric Memory

Removing retrieval lowers quality for both adaptation paradigms. Q_{main} drops from 0.669 to 0.263 for RAFT (RAFT-RAG to RAFT-Closed, a gap of 0.406) and from 0.667 to 0.185 for CLM (CLM-RAG to CLM-Closed, a gap of 0.482). This pattern holds across both S_{det} and S_{asst} : for the CLM system, S_{det} drops from 0.599 to 0.135 and S_{asst} from 0.826 to 0.303. These gaps show that retrieval is not a redundant supplement to parametric adaptation in this setting.

The D2L control (D2L-Closed) supports the same conclusion from a separate engineering path. It reaches $Q_{\text{main}} = 0.210$, slightly above the pure CLM control but far below any retrieval-aware system. Its $S_{\text{asst}} = 0.385$ suggests that the hypernetwork-generated adapter retains some corpus-level language patterns, but without evidence retrieval this is insufficient for factual legal QA. Although the D2L setup differs architecturally from the active CLM setup, it also performs below every retrieval-aware system under the evaluated resource constraints.

These results indicate that retrieval remains the dominant memory mechanism in this setting. Parametric adaptation without evidence access is insufficient, regardless of whether the adapter was trained with supervised QA labels (RAFT-Closed) or corpus-level language modeling (CLM-Closed).

5.5 Single-Document vs. Multi-Document Difficulty

Multi-document questions score lower than single-document questions across all systems.

Table 4 presents the document-scope breakdown.

Table 4: Q_main by document scope (headline and exploratory systems).
Based on 42 single-document and 8 multi-document evaluation questions.

System	Single-doc	Multi-doc	Delta
Base-RAG	0.696	0.310	-0.386
RAFT-RAG	0.694	0.437	-0.257
CLM-RAG	0.722	0.310	-0.412
Merge-RAG	0.718	0.523	-0.195

The RAG baseline and CLM-RAG both drop to Q_main = 0.310 on multi-document items, while RAFT-RAG reaches 0.437 and Merge-RAG reaches 0.523. This breakdown reveals a sharper behavioral distinction than the aggregate table alone.

On single-document questions, CLM-RAG has the highest observed mean at 0.722, narrowly above Merge-RAG at 0.718. Given the 0.004 difference, overlapping seed-level variation, and absence of a paired significance test, this result does not establish a meaningful CLM-RAG advantage. On multi-document questions, CLM-RAG offers no improvement over the RAG baseline (both score 0.310), indicating that the CLM signal does not help with cross-document aggregation.

RAFT-style supervision confers greater robustness to multi-document composition: RAFT-RAG’s multi-doc score (0.437) represents a 41% relative improvement over the baseline’s 0.310. The RAFT training format, which includes distractors alongside gold chunks, may teach the generator to discriminate between relevant and irrelevant evidence, which helps when evidence spans multiple documents.

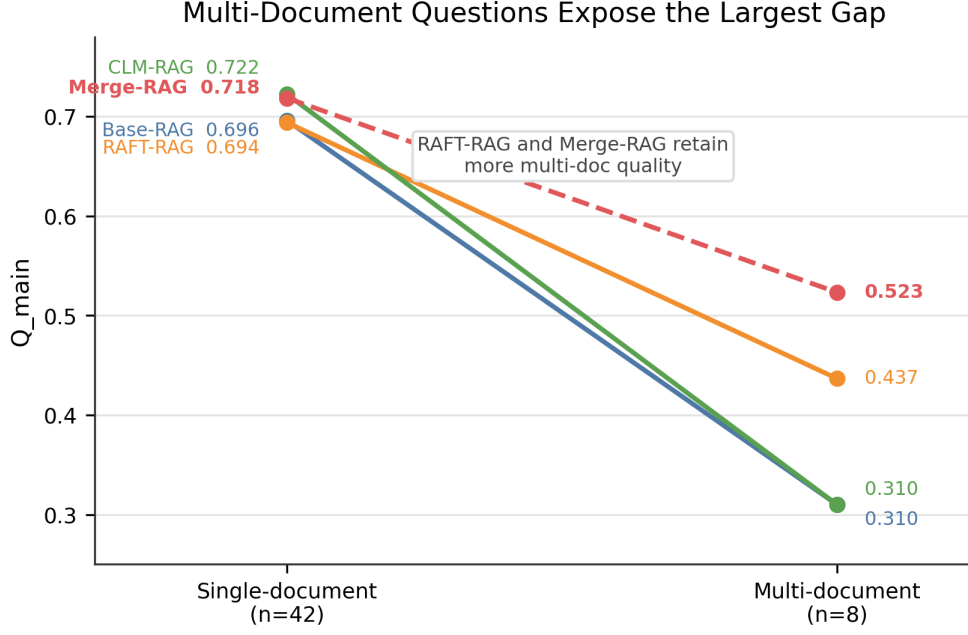


Figure 5: Single-document vs. multi-document Q_{main} per system, annotated with per-system delta.

5.6 Exploratory Adapter Fusion

The merged adapter Merge-RAG provides evidence that the two adaptation signals are not redundant. Relative to RAFT-RAG, Merge-RAG improves Q_{main} by +0.036, S_{det} by +0.031, and S_{asst} by +0.046. Relative to CLM-RAG, it improves Q_{main} by +0.037 and S_{det} by +0.080, while reducing S_{asst} by -0.062. In the document-scope breakdown, Merge-RAG reaches 0.718 on single-document questions, just below CLM-RAG at 0.722, and the highest multi-document score at 0.523. It also has the smallest single-to-multi-document gap (delta = -0.195 vs. -0.386 for the RAG baseline). This pattern is consistent with partial complementarity: the merged system preserves part of the CLM advantage in assistant-style quality while recovering most of the deterministic advantage associated with RAFT-style supervision.

The result is methodologically consistent with recent work on LoRA adapter composition: Prabhakar et al. (2024) show that adapter merge schemes can approach multi-task training quality without retraining, and more structured alternatives such as rank-wise clustering (Zhao et al., 2024) suggest further room for improvement. The current experiment uses a simple linear merge.

The result remains post-hoc for two reasons. First, Merge-RAG is a merge rather than a separately trained system, identified after the main experiments, and its advantage carries higher seed variance (std = 0.035; per-seed Q_main 0.734 / 0.667 / 0.713, the widest spread among the trained retrieval-aware systems). Second, its practical cost is not directly comparable to the headline systems because it inherits prior adaptation cost from both source adapters. The merged system therefore informs interpretation rather than selecting a practical winner, and the headline comparison stands without it. As a post-hoc result rather than a settled one, it supports signal complementarity without establishing a general recipe.

6 Discussion and Limitations

6.1 Answer to RQ1

RQ1 asked whether parametric adaptation yields gains over the RAG baseline and how RAFT-style and CLM adaptation differ. The answer is a qualified affirmative.

Both RAFT-RAG and CLM-RAG improve over the nonparametric RAG baseline, but the observed gains are moderate (+0.026 and +0.025 Q_{main} respectively) and are not accompanied by a paired significance test or confidence interval over the 50 evaluation questions. Because the baseline already reaches $Q_{\text{main}} = 0.643$, modest gains are more informative than they would be against a lower baseline. They indicate that adaptation can still matter after retrieval is in place, but they do not establish a qualitative change in the task.

The choice of training signal proves more consequential than the presence of an adapter per se. RAFT-style supervision improves deterministic extraction (S_{det} : +0.047 over the baseline) at the cost of a slight decrease in free-text quality (S_{asst} : -0.021). CLM continued pretraining improves free-text answer quality (S_{asst} : +0.087) while leaving deterministic extraction essentially unchanged (S_{det} : -0.002). These complementary profiles mean that the optimal system depends on the deployment priority: factual precision favors RAFT-RAG, while explanation quality favors CLM-RAG. CLM-RAG is also roughly half as expensive to train.

The post-hoc merge Merge-RAG achieves the highest aggregate score (0.705), suggesting that the two signals are partially complementary. However, because Merge-RAG was not retrained and was identified post-hoc, this finding should be interpreted as a direction for future work.

6.2 Answer to RQ2

RQ2 asked whether pure parametric systems can substitute for retrieval. The answer is negative within the present setup: retrieval remains necessary.

Neither supervised closed-book adaptation (RAFT-Closed, $Q_{\text{main}} = 0.263$) nor corpus-level CLM pretraining without retrieval (CLM-Closed, $Q_{\text{main}} = 0.185$) provides a viable substitute for external evidence retrieval. The D2L control ($Q_{\text{main}} = 0.210$) corroborates this from a third direction. Under this setup, the 2-billion-parameter model did not internalize enough factual detail to answer legal questions without external evidence.

Within the evaluated corpus, split, model, and hardware regime, retrieval remains the main carrier of document knowledge, while parametric adaptation improves how retrieved evidence is used.

6.3 Error Analysis

Error overlap analysis clarifies both the shared difficulty of the benchmark and the limits of any single system improvement. Fifteen of the 50 evaluation questions are missed by all headline systems (Base-RAG, RAFT-RAG, CLM-RAG, and Merge-RAG). The mean pairwise Jaccard overlap of their failure sets is 0.714, showing that the systems share many errors without identifying whether those errors originate in retrieval coverage, question ambiguity, or generation.

Persistent failure patterns include date extraction (scores at or below 0.400 for all systems), multi-name list normalization (at or below 0.450), and cross-document composition. Among the 15 universally missed questions, recurring themes include unanswerable questions where the gold answer is null, questions requiring information from document regions not well covered by the 3-chunk evidence budget, and questions demanding multi-step cross-document reasoning. Several of these errors persist even when retrieval succeeds, which implies that access to evidence is necessary but not sufficient: some failures reflect remaining difficulty in mapping retrieved context to precise answer behavior.

Local wins by individual systems are sparse: 2 questions are answered correctly only by Base-RAG, 2 only by CLM-RAG, and 0 only by RAFT-RAG or only by Merge-RAG. This limited local complementarity suggests that while the systems have different strengths in aggregate, their per-question advantages rarely translate into exclusive wins, consistent with the modest aggregate deltas observed in Section 5.1.

6.4 Limitations

These findings are bounded in several important respects:

- **Compact corpus.** The benchmark comprises only 8 documents (~115K tokens). Results may not generalize to larger, more heterogeneous corpora; the conclusions should be understood as benchmark-specific and hardware-specific.
- **Small evaluation set.** With 50 evaluation questions, per-type sample sizes are small (as few as $n=5$ for dates and multi-name lists), limiting statistical power for type-level conclusions.
- **Single model.** All experiments use Gemma-2-2b-it. Different model families or scales might alter the relative benefit of parametric adaptation.
- **Training-recipe confounding.** RAFT and CLM differ in objective, data, learning rate, epoch count, and maximum sequence length. Their results compare complete adaptation recipes rather than the training objective alone.
- **Fixed retrieval.** Because retrieval is frozen, the evaluation measures differences in evidence-conditioned generation but cannot assess how adapters interact with retrieval quality or speak to alternative retrieval designs. This strengthens interpretability at the cost of generality.
- **Judge-based free-text scoring.** S_{asst} depends on a frozen judge rubric evaluated by GPT-5.4-mini, introducing potential systematic biases; the manual audit mitigates but does not eliminate this concern.
- **Adapter Fusion Cost.** While Merge-RAG requires no additional training steps to create, its total offline cost necessarily inherits the prior adaptation effort from both

the RAFT and CLM source adapters. Future work could explore whether joint training strategies can achieve similar orthogonal alignment in a single pass.

- **D2L under Resource Constraints.** The observed per-adapter token limit and the cost of training a bespoke Doc-to-LoRA hypernetwork for a modern target model limit its applicability to corpus-scale RAG on resource-constrained hardware. The negative finding is grounded in the token limit and chunk-level workaround, not only in implementation effort.

7 Conclusion

The experiments assessed whether parametric adaptation adds value on top of the RAG baseline for document-grounded legal QA on resource-constrained hardware. The main findings are:

Retrieval remains necessary in this setting. The RAG baseline achieves $Q_{\text{main}} = 0.643$ on the DIFC legal benchmark, while the closed-book controls remain below 0.27. Under this setup, adaptation does not replace retrieval.

Adaptation changes the quality profile. Relative to the RAG baseline, RAFT-style supervision improves deterministic extraction, whereas CLM pretraining improves free-text explanations.

Adapter fusion can partially combine the two profiles. Because the RAFT and CLM signals improve different quality dimensions, their post-hoc linear merge (Merge-RAG) recovers the deterministic advantage while retaining part of the free-text gain. It reaches the highest observed aggregate score (0.705) and the highest multi-document score, where the base model and CLM adaptation perform worse. This result is post-hoc and carries higher seed variance, so it points to a direction rather than a settled recipe.

Under the evaluated resource constraints, retrieval engineering should remain the first priority. Once retrieval is established, targeted adaptation can align generation behavior toward extraction or synthesis. A simple adapter merge combines part of both profiles without an additional training run, but the post-hoc and higher-variance result requires confirmation in a pre-registered, multi-seed evaluation. Future work should examine retrieval-aware adaptation strategies that explicitly target multi-document evidence composition and unanswerable-question calibration.

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A Hyperparameters and Prompts

A.1 QLoRA Configuration (Shared)

Parameter	Value
PEFT method	QLoRA
Rank	32
Alpha	32
Dropout	0.05
Target modules	q_proj, v_proj
Quantization	4-bit NF4, double quantization
Optimizer	Paged AdamW 8-bit
Scheduler	Cosine
Weight decay	0.01

A.2 Training-Signal-Specific Parameters

Parameter	RAFT-RAG	RAFT-Closed (closed-book)	CLM-Closed / CLM-RAG (CLM)
Learning rate	2×10^{-4}	2×10^{-4}	5×10^{-5}
Epochs	3	3	5
Warmup ratio	0.03	0.03	0.10
Max seq. length	4096	4096	512
Effective batch size	4	4 (micro-batch 1, grad. accum. 4)	4 (micro-batch 1, grad. accum. 4)
Training data	150 QA pairs (RAFT format)	150 QA pairs (no context)	~115K tokens (raw corpus)
Supervision	Supervised (question + evidence -> answer)	Supervised (question -> answer)	Unsupervised (next-token)

RAFT-Closed differs from RAFT-RAG only in training data format: retrieved context is omitted. The CLM-Closed/CLM-RAG adapter is trained once via CLM and reused

either without retrieval (CLM-Closed) or with retrieval (CLM-RAG). The CLM maximum sequence length of 512 is a hardware constraint: CLM computes loss over all tokens, and longer sequences exceeded the available memory at the logits stage.

A.3 Retrieval Parameters

Parameter	Value
Embedding model	Qwen3-Embedding-0.6B (384-dim)
Sparse encoder	BM25 Okapi (k1=1.5, b=0.75)
Chunk size (microchunk)	300 tokens
Chunk overlap	50 tokens
Chunk families	page, section, clause, microchunk, table
Candidate prefetch	30 (per channel)
RRF k	60
RRF weights	dense=1.0, sparse=1.0
Post-fusion candidates	10
Reranker	Qwen3-Reranker-0.6B
Rerank budget	5
Evidence budget	3
Max chunks per page	1 (page-diverse)

A.4 Generation Parameters

Parameter	Value
Model	Gemma-2-2b-it
Temperature	0.0 (greedy)
Max new tokens	256
Constrained decoding	Boolean and names types (via Outlines)

A.5 Judge Prompt (Frozen)

System: “You are an impartial judge evaluating a legal QA system’s response. Score each criterion as 1 (met) or 0 (not met). Return ONLY a JSON object.”

User template:

Question: {question}

Reference answer: {reference_answer}

System response: {system_response}

Criteria:

1. correctness: Does the response contain the key information from the reference and no factual errors?
2. completeness: Does the response address all aspects of the question?
3. grounding: Is every claim supported by plausible legal reasoning (no hallucinated specifics)?
4. calibration: Does the response appropriately express uncertainty when information is missing?
5. clarity: Is the answer clear, concise, and directly addresses the question?

Return JSON: {"correctness": 0|1, "completeness": 0|1, "grounding": 0|1, "calibration": 0|1, "clarity": 0|1}

Judge model: GPT-5.4-mini (OpenAI; model id gpt-5.4-mini), reasoning effort = medium, held fixed across all systems and experiments. The prompt is identical for all systems. Malformed judge output is retried once; if the retry also fails, all five criteria are scored as zero for that answer. Judge-based scoring is never applied to deterministic answer types. A manual audit of approximately 10% of judged free-text responses was performed before final interpretation, spot-checking judge scores against the rubric for systematic errors.

B Supplementary Tables and Figures

B.1 Control System Per-Type Breakdown

Table B.1: Per-type scores for control systems. As in Table 3, each cell is the type-appropriate score: S_{det} for the deterministic types and S_{asst} for free-text.

System	Number		Name	Names	Date (n=5)	Free-text (n=13)
	Boolean (n=12)	(n=7)	(n=8)	(n=5)		
RAFT-Closed	0.750	0.143	0.000	0.000	0.000	0.246
CLM-Closed	0.333	0.000	0.125	0.000	0.000	0.303
D2L-Closed	0.333	0.000	0.125	0.000	0.000	0.385

B.2 Seed-Level Variance

Table B.2: Per-seed Q_{main} for trained systems.

Seed	RAFT-RAG	CLM-RAG	Merge-RAG	RAFT-Closed	CLM-Closed
42	0.673	0.651	0.734	0.263	0.182
123	0.654	0.693	0.667	0.258	0.187
777	0.680	0.656	0.713	0.268	0.187
Std	0.014	0.023	0.035	0.005	0.003

B.3 Operational Metrics (All Systems)

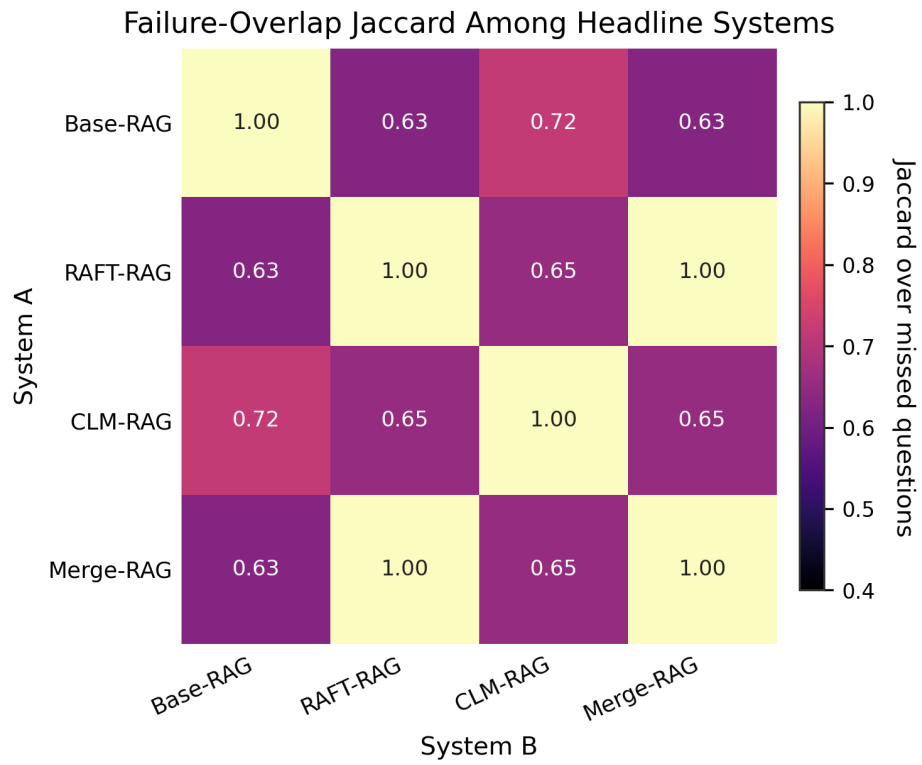


Figure B.1: Failure-overlap Jaccard among headline systems. Higher values indicate that two systems fail on more of the same evaluation questions.

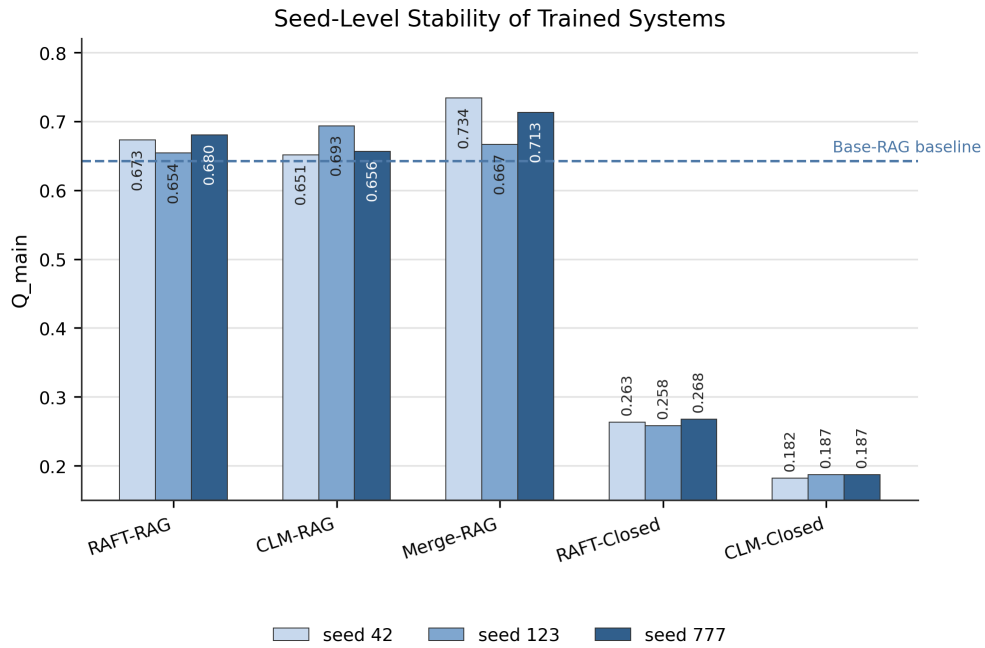
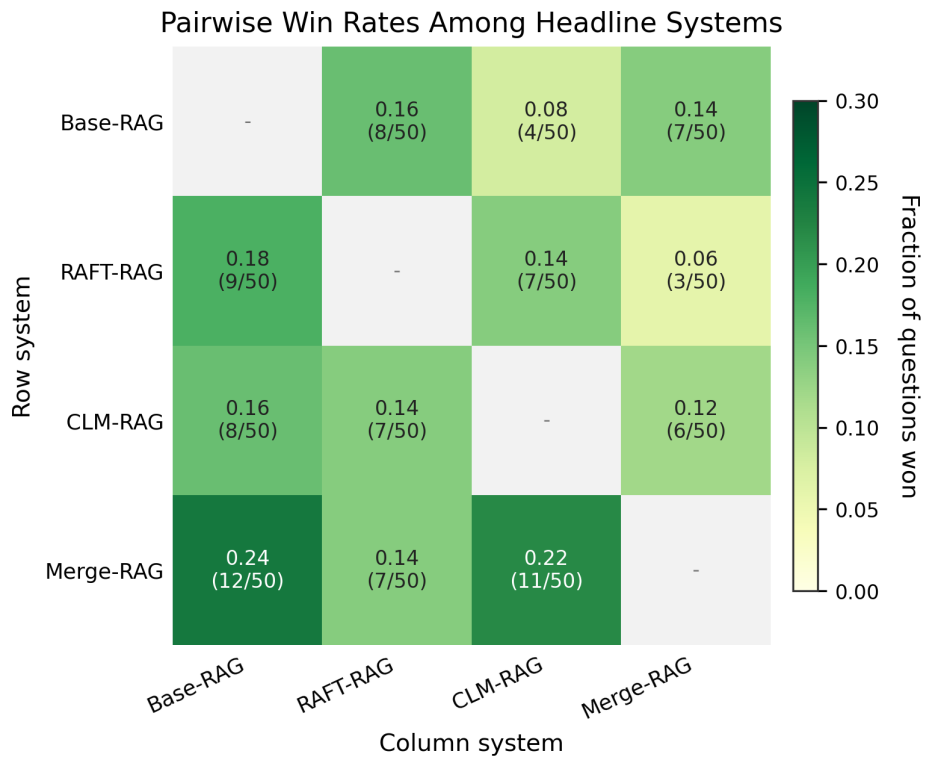


Figure B.2: Per-seed Q_{main} for trained systems, with the Base-RAG baseline shown as a dashed reference line.



Cell = row system scores higher than column system; ties are not wins.

Figure B.3: Pairwise win rates among headline systems. Each off-diagonal cell reports the fraction of evaluation questions where the row system scores higher than the column system; ties are not counted as wins.

Table B.3: Operational metrics on the 50-question evaluation set. TTFT is median time-to-first-token; latency is median and 95th-percentile end-to-end. Peak VRAM is measured at inference. Offline cost is per-seed training wall-clock. Malformed rate is the fraction of evaluation answers that failed structured parsing. Values are medians across seeds for trained systems.

System	TTFT med (ms)	E2E med (ms)	E2E p95 (ms)	Peak infer VRAM (MB)	Offline (s)	Malformed
Base-RAG	335	479	2089	5201	—	0.02
RAFT-RAG	319	492	1966	3069	1206	0.00
CLM-RAG	316	525	2869	3069	581	0.00
Merge-RAG	335	527	1949	3069	n.c.	0.00
RAFT-Closed	51	257	1223	3067	88	0.00
CLM-Closed	58	195	2350	3077	581	0.20
D2L-Closed	56	179	1606	3072	3932	0.16

Malformed rates are negligible for the retrieval-aware adapted systems but rise for the closed-book controls (CLM-Closed 0.20, D2L-Closed 0.16), consistent with their weaker control over structured output formatting without evidence context.

C Doc-to-LoRA Limitations in This Setup

The Doc-to-LoRA (D2L) approach generates document-specific LoRA adapters via a hypernetwork, theoretically allowing a model to internalize context. Its application to RAG corpora on resource-constrained hardware exposes practical limitations.

First, the D2L architecture imposes a strict token limit per generated adapter. A preliminary token-based audit suggested that the 8 documents (8.4K-20.1K D2L context tokens each, ~106K context tokens total) would fit a single-pass D2L encoding, but the released implementation enforced stricter effective limits: every document had to be split into 9-20 chunks (108 chunk-adapters in total), each chunk yielding a separate adapter, with the adapters then merged via linear interpolation. This chunk-level workaround is documented as an engineering diagnostic rather than a strict D2L implementation; it departs from the intended document-level conditioning and added substantial offline cost (3932 s, versus 1206 s for RAFT-RAG training). The per-adapter token limit is tied to the target model’s context window, so RAG-scale corpora exceed it on a compact, context-limited model. A wider context window would raise this ceiling, including for the model on which the released hypernetwork was trained.

Second, applying D2L to a modern target model would require training a bespoke hypernetwork. This was not attempted here; its cost falls outside the evaluated resource budget and remains a design-level consideration rather than a measured result.

The resulting system (D2L-Closed) achieved $Q_{\text{main}} = 0.210$ without retrieval, placing it between the two pure parametric controls (RAFT-Closed = 0.263, CLM-Closed = 0.185) but far below any retrieval-aware system. Under the evaluated hardware budget, the available context window limits how much document content can condition each generated adapter. D2L therefore serves as an engineering diagnostic rather than a competitive alternative to RAFT or CLM in this setting.

D Use of Generative AI

The following generative AI tools were used during the preparation of this work:

- **Claude (Anthropic):** Experiment orchestration, code generation for evaluation and training, and data analysis.
- **GPT-5.4-mini (OpenAI):** Used as the judge model for free-text answer evaluation (S_asst scoring). The judge prompt is reproduced in Appendix A.5.

Responsibility for the final manuscript remains with the author. If the institutional template requires explicit marking of substantially AI-assisted passages, that marking should be applied during the final formatting pass.